



FINAL REPORT

End-to-End Vulnerability Management Cycle

Date: July 2, 2026 | Asset: Metasploitable 2 (10.10.224.235)

Executive Summary

This report documents the complete vulnerability management lifecycle performed on Metasploitable 2, a vulnerable test system. The project demonstrated a comprehensive approach to identifying, prioritizing, remediating, and reporting security vulnerabilities.

Key Achievements

- ✓ Successfully discovered **110 vulnerabilities**
- ✓ Applied **7 manual fixes** (backdoor services, passwords, SSH keys)
- ✓ Reduced open ports from **24 to 12** (50% reduction)
- ✓ Documented **16 exceptions** for EOL OS vulnerabilities
- ✓ Established KPI-driven measurement framework
- ✓ Created comprehensive executive dashboard

Project Summary

Asset Information

Attribute	Value
Asset ID	MSF-01
IP Address	10.10.224.235
Hostname	metasploitable
Operating System	Ubuntu 8.04 LTS (Hardy Heron) - EOL
Environment	VirtualBox Lab (Bridged Network)

Vulnerability Summary

Severity	Count	Percentage	Action
CRITICAL	7	6.4%	Exception Required (EOL)
HIGH	4	3.6%	Exception Required (EOL)
MEDIUM	12	10.9%	Exception Required (EOL)
MIXED/INFO	87	79.1%	Monitor
TOTAL	110	100%	



Risk Posture Improvement

⚠ Important: Ubuntu 8.04 LTS is End-of-Life (May 9, 2013). All official repositories are offline (404 Not Found). System cannot be patched via apt-get. Exceptions have been documented for all package-based vulnerabilities.

Metric	Before	After	Improvement
Risk Score	9.5/10	8.5/10	10.5%
Open Ports	24	12	50%
Backdoor Services	6 running	0 running	100%
Default Passwords	3 (VNC, root, msfadmin)	0	100%
SSLv2/v3 Enabled	Yes	No	Disabled



Deliverables Summary

Week	Deliverable	Status
Week 1	Asset Inventory	✓ Complete
Week 1	Scanning Schedule & Scope	✓ Complete
Week 2	Prioritized Vulnerability Report	✓ Complete
Week 2	CVSS Metrics & Risk Mapping	✓ Complete
Week 3	Remediation Matrix	✓ Complete
Week 3	Exception Documentation	✓ Complete
Week 4	Executive Dashboard & KPIs	✓ Complete
Week 4	Final Report	✓ Complete

Recommendations

Immediate (0-30 Days)

- Document exception for EOL OS vulnerabilities
- Maintain weekly vulnerability scanning
- Monitor for any new critical vulnerabilities
- Keep compensating controls in place

Short-Term (1-3 Months)

- **Replace Ubuntu 8.04 with Ubuntu 22.04 LTS or later**
- Implement automated patch management
- Disable all legacy services (rlogin, rsh, telnet)
- Harden web server configuration

Long-Term (3-6 Months)

- Implement automated vulnerability management platform
- Integrate vulnerability scanning with SIEM
- Establish security awareness training
- Develop incident response playbooks for high-severity findings

Conclusion

This vulnerability management project successfully demonstrated the complete lifecycle:

- **Identify:** 110 vulnerabilities discovered using Nessus
- **Prioritize:** CVSS scoring used to rank vulnerabilities
- **Remediate:** 7 manual fixes applied, 16 exceptions documented
- **Report:** Executive dashboard and KPIs created

Key Takeaway: The system cannot be fully patched due to EOL OS. The recommended course of action is to replace the operating system with a modern, supported version.

